

Schottky Diode & Applications

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I. ABSTRACT

Schottky diodes, known for their unique metal-semiconductor junctions, exhibit distinct characteristics such as low forward voltage drop, fast switching speed, and minimal reverse recovery time. These features make them highly efficient in modern electronics, particularly in power rectification, high-speed switching, and RF applications. This paper explores the physical properties, formation, and operation of Schottky diodes, detailing their DC and AC characteristics with numerical evaluations. Furthermore, it discusses their applications in solar panels, voltage clamping, TTL logic circuits, and more, highlighting their indispensable role in contemporary electronic systems.

II. INTRODUCTION

Schottky diodes are a cornerstone in modern electronics due to their remarkable performance characteristics, which stem from their metal-semiconductor junction. Unlike traditional PN junction diodes, Schottky diodes offer a significantly lower forward voltage drop, enabling enhanced energy efficiency and reduced power losses. Additionally, their high-speed switching capabilities make them ideal for use in applications requiring rapid response times, such as digital circuits and RF systems.

The versatility of Schottky diodes extends to numerous domains, from power management systems and solar energy protection to signal demodulation and high-frequency communication systems. However, their unique properties, such as higher reverse leakage current and limited breakdown voltage, necessitate careful consideration in design and application.

This paper aims to provide a comprehensive overview of Schottky diodes, delving into their structure, operational principles, and electrical characteristics. Through detailed numerical evaluations and graphical analyses, it illustrates the key factors influencing their performance. Furthermore, it highlights the pivotal roles of Schottky diodes in diverse applications, solidifying their status as an essential component in modern technology.

III. STRUCTURE

A. Physical Properties

Schottky diodes are characterized by their unique metal-semiconductor junction, which imparts several distinct physical properties [1]:

1) Structure

- It is composed of a metal-semiconductor junction rather than a PN junction. Metal: Commonly platinum, aluminum, gold, or molybdenum. Semiconductor: Typically, N-type silicon, though gallium arsenide or silicon carbide may be used for specialized applications.

2) Low Forward Voltage Drop

- Forward voltage drop ranges from 0.2V to 0.4V, reducing power losses compared to PN junction diodes (~0.7V).

3) Reverse Leakage Current

- Exhibits higher reverse leakage current than standard diodes, which increases with temperature.

4) High-Speed Switching

- Minimal minority carrier storage ensures rapid switching and is suitable for high-frequency applications.

5) Thermal Stability and Breakdown Voltage

- Thermal conductivity of the metal enhances heat dissipation.
- Typical breakdown voltage ranges from 20V to 200V, lower than PN junction diodes.

6) Capacitance

- Lower capacitance makes Schottky diodes advantageous in RF and high-frequency circuits.

B. Formation of Schottky Diodes

Schottky diodes are formed through the precise creation of a metal-semiconductor junction [2]:

1) Metal-Semiconductor Junction

- A Schottky diode consists of a metal (e.g., platinum, gold, molybdenum) in contact with an N-type semiconductor (e.g., silicon).
- The work function difference between the metal and semiconductor creates a Schottky barrier.

2) Formation of the Schottky Barrier

- Electrons flow from the semiconductor to the metal upon contact, creating a depletion region on the semiconductor side.
- This region establishes the Schottky barrier, controlling current flow.

- The barrier height ϕ_B can be approximated as: $\{\phi_B = \phi_m - x(1)\}$ where ϕ_m is the metal work function and is the electron affinity of the semiconductor.

3) Ohmic Contact

- The opposite side of the semiconductor is equipped with a low-resistance ohmic contact to facilitate efficient current flow

4) Manufacturing Process

- Surface Preparation: The semiconductor surface is cleaned and polished to ensure defect-free junctions.
- Metal Deposition: Techniques like sputtering or evaporation are used to deposit the metal layer.
- Encapsulation: The diode is encapsulated to protect against environmental factors

IV. BASIC IDEA OF OPERATION

To demonstrate the operation of Schottky diodes, it is essential to first consider their behavior in the unbiased state. In this condition, Schottky diodes maintain thermal equilibrium, allowing for a detailed examination of their fundamental characteristics and underlying principles [3]:

1) Energy Barrier Formation

- The difference in work functions between the metal and semiconductor creates a potential barrier at the junction.
- Electrons in the semiconductor align with the metal's Fermi level, forming the Schottky barrier.

2) Depletion Region

- A thin depletion region forms on the semiconductor side, containing immobile positively charged donor ions.
- The depletion width W can be calculated as

$$\sqrt{\frac{2\epsilon_s(V_B - V)}{qN_D}}$$

3) Thermal Equilibrium

- Electron flow from the semiconductor to the metal is balanced by reverse electron flow, resulting in no net current flow.

4) Barrier Height

- The Schottky barrier height is determined by the metal's work function and the semiconductor's electron affinity (typically 0.6 eV to 0.8 eV)
- Current density J in the diode can be expressed as

$$J = A^* T^2 \exp\left(\frac{\phi_B}{kT}\right)$$

where A^* is the Richardson constant, T is the temperature, ϕ_B is the barrier height, and k is Boltzmann's constant.

5) No Minority Carrier Involvement

- Schottky diodes do not rely on minority carrier recombination, enabling faster response times.

V. LARGE CHARACTERISTICS AND NUMERICAL EVALUATION

A. DC Bias

The forward current-voltage (I-V) characteristics [4] of a Schottky diode obeying the thermionic emission model are given by

$$I = I_S \left[\exp\left(\frac{qV_d}{kT}\right) - 1 \right] \quad (1)$$

Most practical Schottky diodes show deviations from ideal thermionic emission behavior. A dimensionless parameter called the ideality factor, n , is usually included in the I-V relationship to take into account nonideal diode behaviors

$$I = I_S \left[\exp\left(\frac{qV_d}{nkT}\right) - 1 \right] \quad (2)$$

where q is the electronic charge, V_d the voltage applied across the diode, k the Boltzmann constant, and T the absolute temperature. I_s can be expressed by

$$I_s = A^* A_{eff} T^2 \exp\left(\frac{-q\phi_B}{kT}\right) \quad (3)$$

Where A_{eff} is the effective area of the diode, A^* is the Richardson constant (equal to $32 \text{ A/cm}^2 \text{K}^2$ for p-type Si), and ϕ_B is the Schottky barrier height of the diode.

The effect of the diode series resistance R is usually modeled with a series combination of a diode and a resistor with resistance R through which the current I flows. The voltage V_d across the diode can then be expressed in terms of the total voltage drop V across the series combination of the diode and the resistor. Thus, $V_d = V - IR$, and for $V_d \gg \frac{3kT}{q}$ Eq. 2 becomes

$$I = I_S \left[\exp\left(\frac{q(V - IR)}{nkT}\right) - 1 \right] \quad (4)$$

The ideality factor n is a measure of the conformity of the diode to pure thermionic emission. The ideality factor is calculated from the slope of the linear region of the forward bias $\ln(IV)$ plot and can be written from Eq. 2 as

$$n = \frac{q}{kT} \frac{dV}{d(\ln I)} \quad (5)$$

Graph (1) shows the forward and reverse bias $\ln(I-V)$ characteristics of the MIS Schottky diode at room temperature. As can be seen from this figure, the (I-VI) characteristic of the MIS Schottky diode shows a good rectifying behaviour with relatively low leakage current of $0.25 \mu\text{A}$ at reverse bias of $V_R = -3\text{V}$. The current curve in forward bias quickly becomes dominated by series resistance from contact wires or bulk resistance of the semiconductor, giving rise to the curvature at high current in the semilogarithmic I-V plot. Using Eqs. (3) and (5), the values of the barrier height and the ideality factor were found to be $\phi_B = 0.786 \text{ eV}$ and $n = 1.766$ respectively.

The ideality factor of the MIS Schottky diode, obtained from the forward bias I-V characteristics, is observed to be greater than unity. This behavior suggests the presence of a thin interfacial insulator layer between the Al layer

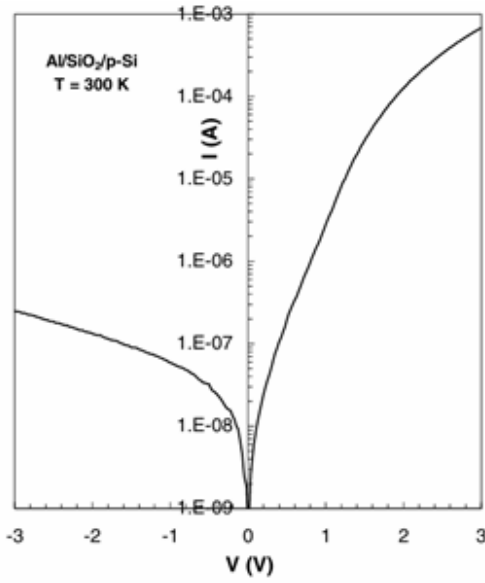


Fig. 1: The forward and reverse bias I-V characteristics of the MIS Schottky diode at room temperature [5]

and the P-Si semiconductor interface. This insulating layer may form due to the exposure of P-Si to air prior to the deposition of the Al layer. Additionally, the observed ideality factor behavior can be attributed to a distribution of interface states, the image-force effect, and recombination-generation processes. Other potential mechanisms, such as tunneling, could also contribute to the observed ideality factor greater than unity. It is important to note that the influence of series resistance in the linear region is negligible in the presented graph. 1

The Contact Resistance A marginal contact is identified by its contact resistance, which is defined as $R_c = (\frac{\partial V}{\partial J})_{V=0}$. R_c is related to the barrier height by Eq. 6

$$R_c = \frac{k}{eA^*T} \exp\left(\frac{e\phi_B}{kT}\right) \quad (6)$$

At high forward bias, the series resistance of a Schottky diode becomes a dominant factor influencing its current-voltage (I-V) characteristics. Series resistance arises due to the resistance of the semiconductor bulk, the metal-semiconductor contact, and external contacts such as wires or interconnects. As the current increases, the voltage drop across this resistance becomes significant, causing deviations from the ideal exponential behavior predicted by the thermionic emission model. This effect leads to a noticeable curvature in the forward bias I-V plot, especially at higher currents, where the total voltage across the diode can be expressed as the sum of the ideal diode voltage and the series resistance contribution.

B. AC Bias

Capacitance-voltage-frequency (C-V-f) characteristics [5] : In Schottky diodes the depletion layer capacitance can be expressed as

$$\frac{1}{C^2} = \frac{2(V_R) + V_I}{A^2 q \epsilon_s N_D} \quad (7)$$

where A is the area of the diode, ϵ_s is the permittivity of semiconductor (Si), N_D is the carrier (acceptor) concentration, q is the electron charge, V_I is the diffusion potential at zero bias, V_R is the magnitude of the reverse bias.

The swept (sweep rate 30 mV/sec) 1 MHz (C-V) characteristics of the MIS (metal insulator semiconductor) and MS (metal semiconductor) diodes are shown in Fig 2. The zero-bias capacitance is seen to drop from 220 to 154 pF due to the interfacial layer. As the reverse voltage is increased the difference in capacitance between the two devices decreases, but it is appreciable even at the highest applied voltage. The forward capacitance was obtained up to the voltage at which conductance becomes significant, and as seen in Fig 2 both devices exhibit the expected increase in capacitance with forward voltage. In order to assess the doping concentration and barrier height, $\frac{1}{C^2}$ vs V_R plots were obtained from the (C-V) data of 2.

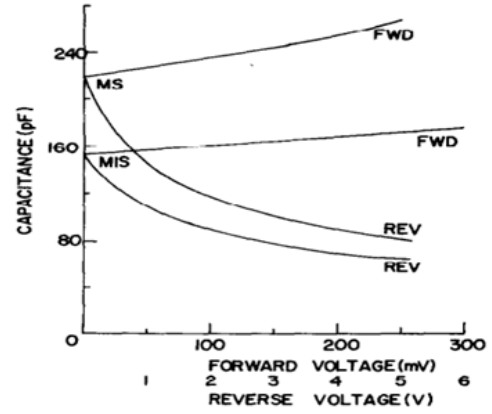


Fig. 2: Swept 1 MHz C-V characteristics of MIS and MS diodes [6]

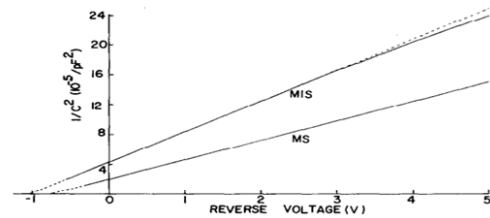


Fig. 3: $\frac{1}{C^2}$ against voltage for MIS and MS diodes (solid line is the data, dotted line above 3V shows slight deviation for MIS diode)

The capacitance–voltage–frequency (C–V–f) measurements were performed at frequencies 100, 200, 500, 700 kHz and 1 MHz by using HP 4192A Impedance Analyser at room temperature.

Graph 4 shows the forward and reverse-bias (C–V) characteristics of MIS Schottky diode at room temperature. As can be seen in Fig 4 the measured capacitance (C) is dependent on bias voltage and frequency. Each (C–V) curves have three regimes of accumulation–depletion–inversion region, but with a considerable voltage-axis shift due to the presence of surface states.

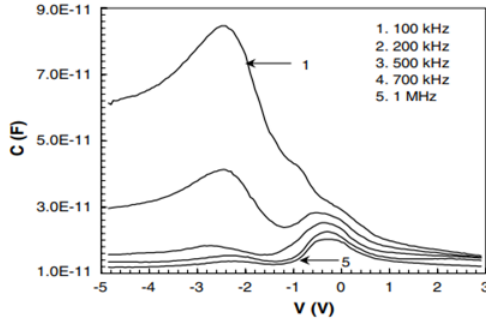


Fig. 4: The forward and reverse bias C–V characteristics of the MIS Schottky diode at 300°K

At high frequency, the MIS Schottky [5] diode with high series resistance showed decreasing capacitance with increasing frequencies. The voltage and frequency dependence is due to the particular features of Schottky barrier, impurity level, high series resistance, doping density, etc. This occurs because at lower frequencies the interface states can follow the ac signal and yield an excess capacitance, which depends on the frequency. In the high frequency limit ($f \geq 500$ kHz), the interface states cannot follow the ac signal. This makes the contribution of interface state capacitance to the total capacitance negligibly small. From the above discussion it can be concluded that under bias (V) the interface states are responsible for the observed frequency dispersion in C–V curves. When an ac signal is applied, there is a frequency dependence of the C–V curve, which does not exist at the ideal case. In this case –V plot is linear but low frequencies C^2 –V plot is not linear. This non-linearity explained as some of the interface states follow the applied ac signal. The C^2 –V plots, which are carried out for various selected frequencies (100 kHz to 1 MHz) is presented in Fig 5. As shown in Fig 5 the C^2 –V plots are linear in wide voltage region when C–V measurements are carried out at sufficiently high frequency (such that carrier life time τ is much larger than $\frac{1}{2\pi f}$). It can be seen that at high frequencies, while the values of the capacitance are low, at lower frequencies these values become higher. This is attributed to the interface states, which cannot follow ac signal at high frequencies.

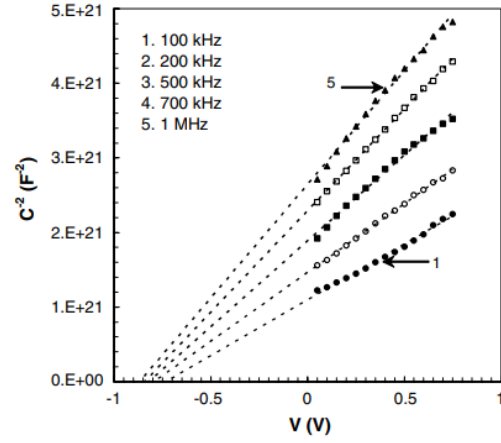


Fig. 5: The reverse bias C^2 –V characteristics of the MIS Schottky diode at different frequencies

VI. APPLICATIONS OF SCHOTTKY DIODES

Schottky diodes are indispensable in modern electronics due to their unique properties such as low forward voltage drop, fast switching speed, and low reverse recovery time. Below is a review of their key applications:

• High-Speed Switching Circuits

- Schottky diodes are extensively used in high-speed switching applications, particularly in digital circuits and power supplies. Their negligible charge storage enables faster operation compared to standard diodes, making them ideal for pulse shaping and clock signal generation.

• Power Rectifiers

- The low forward voltage drop of Schottky diodes results in reduced power loss, which is critical in low-voltage, high-current applications like DC-DC converters and AC-DC adapters. They enhance energy efficiency and are widely used in power management systems.

• RF and Microwave Circuits

- In radio frequency (RF) and microwave systems, Schottky diodes are employed as mixers and detectors. Their high switching speed and low junction capacitance ensure accurate signal processing in applications like radar systems, satellite communications, and wireless transceivers.

• Solar Panel Protection

- Schottky diodes are used as bypass diodes in photovoltaic (PV) systems to prevent reverse current flow that could damage solar cells. Their efficiency and reliability ensure the safe operation of solar panels under varying conditions, such as shading or partial obstruction.

• Signal Demodulation

- In communication systems, Schottky diodes are often used for signal detection and demodulation, particularly in AM and FM receivers. Their fast response time allows them to accurately process high-frequency signals.
- **Voltage Clamping and ESD Protection**
 - Due to their low forward voltage characteristics, Schottky diodes are commonly used in clamping circuits to protect sensitive electronic components from voltage spikes. They are also employed in electrostatic discharge (ESD) protection circuits, ensuring the longevity of integrated circuits (ICs).
- **Battery-Powered Devices**
 - Schottky diodes are used in battery-operated systems such as smartphones and wearable devices to minimize power loss, extend battery life, and improve overall energy efficiency.
- **TTL Logic Circuits**
 - In transistor-transistor logic (TTL) circuits, Schottky diodes are integrated to improve switching speeds by preventing transistor saturation. This enhancement reduces delay and ensures faster performance in digital logic systems.

VII. ADVANTAGES AND DISADVANTAGES [7], [8]

A. *Advantages*

- 1) **Very low V_f**
 - the Schottky diodes consist of metal-semiconductor junction which creates a depletion region without need of recombination of charge carriers which is slower in pn diodes.
 - The range of barrier potential is between 0.2V and 0.48V and this range is lower than that of pn junction diodes .
 - The Schottky diodes are majority carrier devices where the minority carriers effect is weak and this decreases the energy loss with recombination and storage which lowers the V_f .
- 2) **Extremely low switching losses**
 - due to the difference between their minority carriers conduction mechanism with the pn junction diodes, where they don't have reverse recovering time or significant stored charges , this leads to instantaneous transitions between ON and OFF states and this is important in high frequency rectification applications and the switching converters.
- 3) **Low noise switching**
 - Schottky diodes have lower switching noise compared to pn-junction diodes due to their fast response and absence of recovery currents, and this is useful in various applications as RF circuits and communication systems .

4) **High reliability circuit operation**

- Schottky diodes are highly reliable as they don't suffer from reverse recovery issues, which makes them ideal for applications demanding high reliability and long life time as their low forward voltage and resistance to thermal stresses which makes them preferable in power supply designs and high frequency switching circuits and converters , they also have faster switching capability and low power losses.

5) **Low voltage peaks for reduced circuit protections**

- Schottky diodes generate low voltage peaks on transitions between ON and OFF states as a result of absence of reverse recovery currents and instantaneous switching and these low peaks result in reduced voltage transients and spikes during voltage operations and this leads to reducing the need of additional protection components leading to improvement of the capability , efficiency and simplicity of circuits design.

B. *Disadvantages*

1) **Lower breakdown voltage**

- the reverse breakdown voltage of Schottky diodes is less than that of pn-junction diodes due to the metal-semiconductor junction's characteristics of Schottky diodes which limits the maximum reverse voltage which the diode can withstand leading to the difficulty of using these diodes in application which depend in its operation on high voltage like high voltage rectifiers and power transmission systems, so the Schottky diodes are used in more limited voltage applications .

2) **Temperature sensitivity**

- the Schottky diodes are more sensitive to temperature than the pn-junction diodes as the reverse leakage current in Schottky diodes increases exponentially with temperature

3) **Potential for increased electromagnetic interference (EMI)**

- electromagnetic interference is generated as a result of the rapid switching characteristics of the Schottky diode and this leads to high frequency noise which leads to interference with the nearby systems, this problem happens for example the RF communication systems and in the high speed digital circuits

4) **Cost**

- as the manufacturing of Schottky diodes requires the usage of high quality metal-semiconductor junctions, which increase their cost effectively and that may result in the avoidance of purchasing them due to their high cost

VIII. CONCLUSION

Schottky diodes, with their unique metal-semiconductor junctions, have revolutionized the design and functionality of electronic devices. Their inherent advantages, such as low forward voltage drop, fast switching, and reduced power losses, make them indispensable in applications ranging from power rectification to RF systems and voltage clamping. While challenges like higher reverse leakage current and lower breakdown voltage exist, they are mitigated through careful design and material selection.

This paper has comprehensively analyzed the physical properties, operational principles, and numerical characteristics of Schottky diodes, emphasizing their performance and reliability. The discussion on applications further underscores their pivotal role in advancing modern electronics. As technology continues to evolve, the demand for efficient and high-speed components like Schottky diodes will undoubtedly grow, ensuring their sustained significance in the electronic landscape.

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